A Monero data model and API specification

with implementations in Java, TypeScript, and C++
using monero v0.18.5.4 'Flourine Fermi'

Last updated October 22, 2026



Java implementation: https://github.com/woodser/monero-java

TypeScript implementation: https://github.com/woodser/monero-ts

C++ implementation: https://github.com/woodser/monero-cpp

Document source: https://github.com/woodser/monero-java/blob/master/docs/monero-spec.drawio

Document created and revisable using: https://www.draw.io/

Monero Daemon Interface 1/2

setProxyUri(string proxyUri): void

isConnected(): bool

getVersion(): MoneroVersion

isTrusted(): bool

getHeight(): ulong

getBlockHash(ulong height): string

getBlockTemplate(string walletAddress, uint reserveSize=null): MoneroBlockTemplate

getLastBlockHeader(): MoneroBlockHeader

getBlockHeaderByHash(string blockHash): MoneroBlockHeader

getBlockHeaderByHeight(ulong height): MoneroBlockHeader

getBlockHeadersByRange(ulong startHeight=0, ulong endHeight=chainHeight): MoneroBlockHeader[]

getBlockByHash(string blockHash): MoneroBlock

getBlocksByHash(string[] blockHashes, ulong startHeight, bool prune=false): MoneroBlock[]

getBlockByHeight(ulong height): MoneroBlock

getBlocksByHeight(ulong[] heights): MoneroBlock[]

getBlocksByRange(ulong startHeight=0, ulong endHeight=chainHeight): MoneroBlock[]

getBlocksByRangeChunked(ulong startHeight=0, ulong endHeight=chainHeight, ulong maxChunkSize=3000000): MoneroBlock[]

getBlockHashes(string[] blockHashes, ulong startHeight): string[]

getTx(string txHash, bool prune=false): MoneroTx

getTxs(string[] txHashes, bool prune=false): MoneroTx[]

getTxHex(string txHash, bool prune=false): string

getTxHexes(string[] txHashes, bool prune=false): string[]

getMinerTxSum(ulong height, ulong numBlocks=chainHeight): MoneroMinerTxSum

getFeeEstimate(ulong graceBlocks=?): MoneroFeeEstimate

submitTxHex(string txHex, bool doNotRelay=false): MoneroSubmitTxResult

relayTxByHash(string txHash): void

relayTxsByHash(string[] txHashes): void

getTxPool(): MoneroTx[]

getTxPoolHashes(): string[]

getTxPoolBacklog(): MoneroTxBacklogEntry[]

getTxPoolStats(): MoneroTxPoolStats

flushTxPool(string[] txHashes=null): void

getKeyImageSpentStatus(string keyImage): MoneroKeyImageSpentStatus

getKeyImageSpentStatuses(string[] keyImages): MoneroKeyImageSpentStatus[]

getOutputs(MoneroOutput[] outputs): MoneroOutput[]

getInfo(): MoneroDaemonInfo

getSyncInfo(): MoneroDaemonSyncInfo

getHardForkInfo(): MoneroHardForkInfo

Monero Daemon Interface 2/2

getAltChains(): MoneroAltChain[]

getAltBlockHashes(): string[]

getDownloadLimit(): uint

setDowloadLimit(uint limit): uint

resetDownloadLimit(): uint

getUploadLimit(): uint

setUploadLimit(uint): uint

resetUploadLimit(): uint

getPeers(): MoneroPeer[]

getKnownPeers(): MoneroPeer[]

setOutgoingPeerLimit(uint limit): void

setIncomingPeerLimit(uint limit): void

getPeerBans(): MoneroBan[]

setPeerBan(MoneroBan ban): void

setPeerBans(MoneroBan[] bans): void

getOutputDistribution(ulong[] amounts, bool isCumulative, ulong startHeight=0, ulong endHeight=chainHeight):
MoneroOutputDistributionEntry[]

getOutputHistogram(ulong[] amounts, ulong minCount, ulong maxCount, bool isUnlocked, ulong recentCutoff):
MoneroOutputHistogramEntry[]

startMining(string address, ulong numThreads=null, bool isBackground=false, bool ignoreBattery=false): void

stopMining(): void

getMiningStatus(): MoneroMiningStatus

submitBlock(string blockBlob): void

submitBlocks(string[] blockBlobs): void

checkForUpdate(): MoneroDaemonUpdateCheckResult

downloadUpdate(string path): MoneroDaemonUpdateDownloadResult

addListener(MoneroDaemonListener listener): void

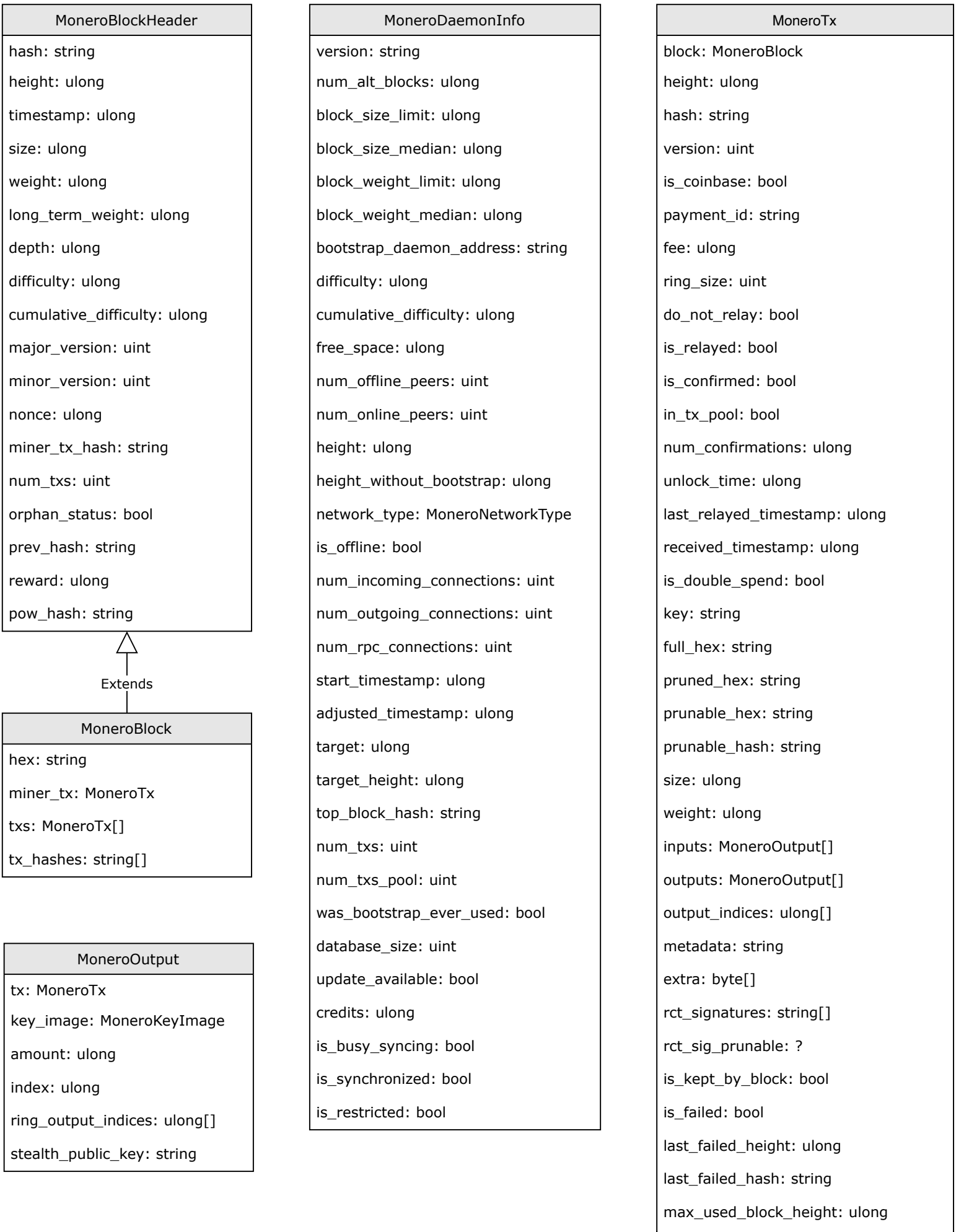
removeListener(MoneroDaemonListener listener): void

waitForNextBlockHeader(): MoneroBlockHeader

stop(): void

pruneBlockchain(bool check): MoneroPruneResult

Monero Daemon Types 1/3



Monero Daemon Types 2/3

MoneroDaemonSyncInfo
height: ulong
connections: MoneroDaemonConnection[]
spans: MoneroDaemonConnectionSpan[]
target_height: ulong
next_needed_pruning_seed: uint
overview: ?

MoneroDaemonListener
onBlockHeader(MoneroBlockHeader header): void

<enumeration> MoneroNetworkType
mainnet: 0
testnet: 1
stagenet: 2

MoneroPeer
id: string
address: string
host: string
port: uint
is_online: boolean
last_seen_timestamp: ulong
pruning_seed: uint
rpc_port: uint
rpc_credits_per_hash: ulong
type: ConnectionType
num_support_flags: uint
state: string
send_idle_time: ulong
receive_idle_time: ulong
num_sends: uint
num_receives: uint
is_local_host: bool
is_local_ip: bool
live_time: ulong
is_incoming: bool
height: ulong
current_upload: ulong
current_download: ulong
avg_upload: ulong
avg_download: ulong

MoneroBan
host: string
ip: string
is_banned: bool
seconds: ulong

MoneroKeyImage
hex: string
signature: string

<enumeration> MoneroKeyImageSpentStatus
not_spent: 0
confirmed: 1
tx_pool: 2

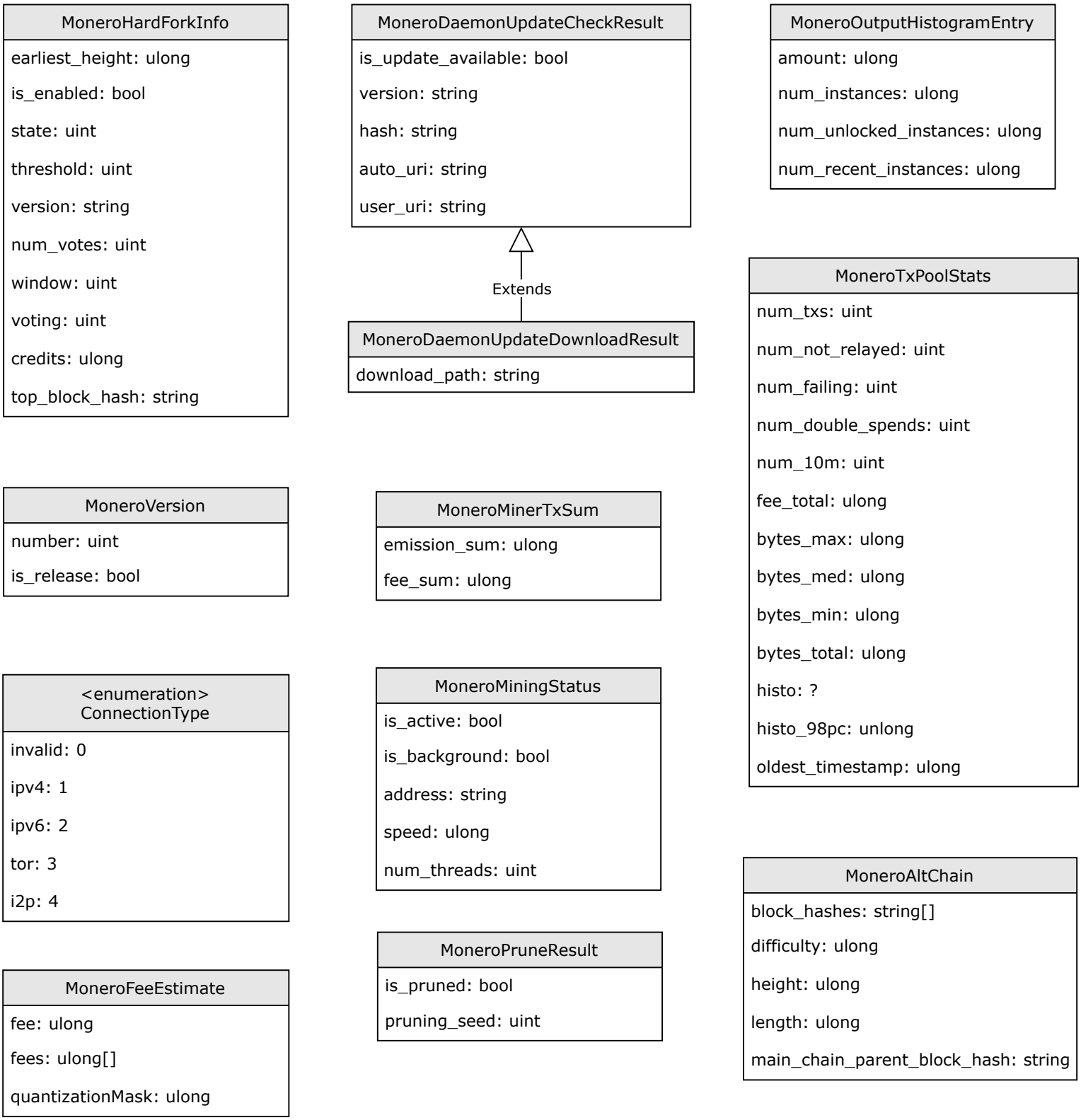
MoneroBlockTemplate
block_template_blob: string
block_hashing_blob: string
difficulty: ulong
expected_reward: ulong
height: ulong
prev_hash: string
reserved_offset: ulong
seed_height: ulong
seed_hash: string
next_seed_hash: string

MoneroSubmitTxResult
is_good: bool
is_relayed: bool
is_double_spend_seen: bool
is_fee_too_low: bool
is_mixin_too_low: bool
has_invalid_input: bool
has_invalid_output: bool
has_too_few_outputs: bool
is_rct: bool
is_overspend: bool
is_too_big: bool
sanity_check_failed: bool
reason: string
credits: ulong
top_block_hash: string
is_tx_result_too_big: bool
nonzero_unlock_time: bool

MoneroDaemonConnectionSpan
connection_id: string
num_blocks: ulong
remote_address: string
rate: ulong
speed: ulong
size: ulong
start_block_height: ulong

max_used_block_hash: string
signatures: string[]

Monero Daemon Types 3/3



Monero Wallet Interface 1/3

addListener(MoneroWalletListener listener): void

removeListener(MoneroWalletListener): void

getListeners(): MoneroWalletListener[]

isViewOnly(): bool

setDaemonConnection(MoneroDaemonConnection connection): void

getDaemonConnection(): MoneroDaemonConnection

isConnectedToDaemon(): bool

getVersion(): MoneroVersion

getNetworkType(): MoneroNetworkType

getPath(): string

getSeed(): string

getSeedLanguage(): string

getSeedLanguages(): string[]

getPrivateSpendKey(): string

getPrivateViewKey(): string

getPublicSpendKey(): string

getPublicViewKey(): string

getPrimaryAddress(): string

getAddress(uint accountIdx, uint subaddressIdx): string

getAddressIndex(string address): MoneroSubaddress

getDaemonConnection(): MoneroRpcConnection

getDaemonHeight(): ulong

getDaemonMaxPeerHeight(): ulong

getApproximateChainHeight(): ulong

getHeight(): ulong

getHeightByDate(uint year, uint month, uint day): ulong

sync(ulong startHeight=null, MoneroSyncListener listener=null): MoneroSyncResult

startSyncing(): void

stopSyncing(): void

scanTxs(string[] txHashes): void

rescanSpent(): void

rescanBlockchain(): void

getBalance(uint accountIdx=null, uint subaddressIdx=null): ulong

getUnlockedBalance(uint accountIdx=null, uint subaddressIdx=null): ulong

getAccounts(bool includeSubaddresses=false, string tag=null): MoneroAccount[]

getAccount(uint accountIdx, bool includeSubaddresses=false): MoneroAccount

createAccount(string label=null): void

getSubaddresses(uint accountIdx, uint[] subaddressIndices): void

getSubaddress(uint accountIdx, uint subaddressIdx): MoneroSubaddress

Monero Wallet Interface 2/3

createSubaddress(uint accountIdx, string label=null): MoneroSubaddress

setAccountLabel(uint accountIdx, string label): void

setSubaddressLabel(uint accountIdx, string subaddressIdx, string label): void

getTx(string txId): MoneroTxWallet

getTxs(MoneroTxQuery query=null): MoneroTxWallet[]

getTransfers(uint accountIdx=null, uint subaddressIdx=null): MoneroTransfer[]

getOutputs(MoneroOutputQuery query=null): MoneroOutputWallet[]

getOutgoingTransfers(MoneroTransferQuery query=null): MoneroOutgoingTransfer[]

getIncomingTransfers(MoneroTransferQuery query=null): MoneroIncomingTransfer[]

getTransfers(MoneroTransferQuery query=null): void

exportOutputs(bool all=false): string

importOutputs(string outputsHex): uint

exportKeyImages(bool all=false): MoneroKeyImageExportResult

importKeyImages(MoneroKeyImage[] keyImages, uint offset=0): MoneroKeyImageImportResult

freezeOutput(String keyImage): void

thawOutput(String keyImage): void

isOutputFrozen(String keyImage): bool

getNewKeyImagesFromLastImport(): MoneroKeyImage[]

getDefaultFeePriority(): MoneroTxPriority

createTx(MoneroTxConfig config): MoneroTxWallet

createTxs(MoneroTxConfig config): MoneroTxWallet[]

sweepOutput(MoneroTxConfig config): MoneroTxWallet

sweepUnlocked(MoneroTxConfig config): MoneroTxWallet[]

sweepDust(bool relay=false): MoneroTxWallet[]

relayTx(string txMetadata): string

relayTx(MoneroTxWallet tx): string

relayTxs(string[] txMetadatas): string[]

relayTxs(MoneroTxWallet[] txs): string[]

describeTxSet(MoneroTxSet txSet): MoneroTxSet

signTxs(string unsignedTxHex): MoneroTxSet

submitTxs(string signedTxHex): string[]

signMessage(string message, MoneroMessageSignatureType type, int accountIdx, int subaddressIdx): string

verifyMessage(string message, string address, string signature): MoneroMessageSignatureResult

getTxKey(string txId): string

checkTxKey(string txId, string txKey, string address): MoneroCheckTx

getTxProof(String txId, string address, string message=null): string

checkTxProof(string txId, string address, string message, string signature): MoneroCheckTx

getSpendProof(string txId, string message=null): string

checkSpendProof(string txId, string message, string signature): bool

Monero Wallet Interface 3/3

getReserveProofWallet(string message): string

getReserveProofAccount(uint accountIdx, ulong amount, string message): string

checkReserveProof(string address, string message, string signature): MoneroCheckReserve

setTxNotes(string[] txHashes, string[] notes): void

setTxNote(string txHash, string note): void

getTxNotes(string[] txHashes): string[]

getTxNote(string txHash): string

getAddressBookEntries(uint[] entryIndices=null): MoneroAddressBookEntry[]

addAddressBookEntry(string address, string description): uint

editAddressBookEntry(uint entryIdx, bool setAddress, string address, bool setDescription, string description): void

deleteAddressBookEntry(uint entryIdx): void

tagAccounts(string tag, uint[] accountIndices): void

untagAccounts(uint[] accountIndices): void

getAccountTags(): MoneroAccountTag[]

setAccountTagLabel(string tag, string label): void

getPaymentUri(MoneroTxConfig request): string

parsePaymentUri(string uri): MoneroSendRequest

getAttribute(string key): string

setAttribute(string key, string val): void

startMining(uint numThreads=null, bool backgroundMining=false, bool ignoreBattery=true): void

stopMining(): void

isMultisigImportNeeded(): bool

isMultisig(): bool

getMultisigInfo(): MoneroMultisigInfo

prepareMultisig(): string

makeMultisig(string[] multisigHexes, uint threshold, string password): string

exchangeMultisigKeys(string[] multisigHexes, string password): MoneroMultisigInitResult

exportMultisigHex(): string

importMultisigHex(string[] multisigHexes): uint

signMultisigTxHex(string multsiigTxHex): MoneroMultisigSignResult

submitMultisig(string signedMultisigHex): string[]

moveTo(string path): void

changePassword(string oldPassword, string newPassword): void

save(): void

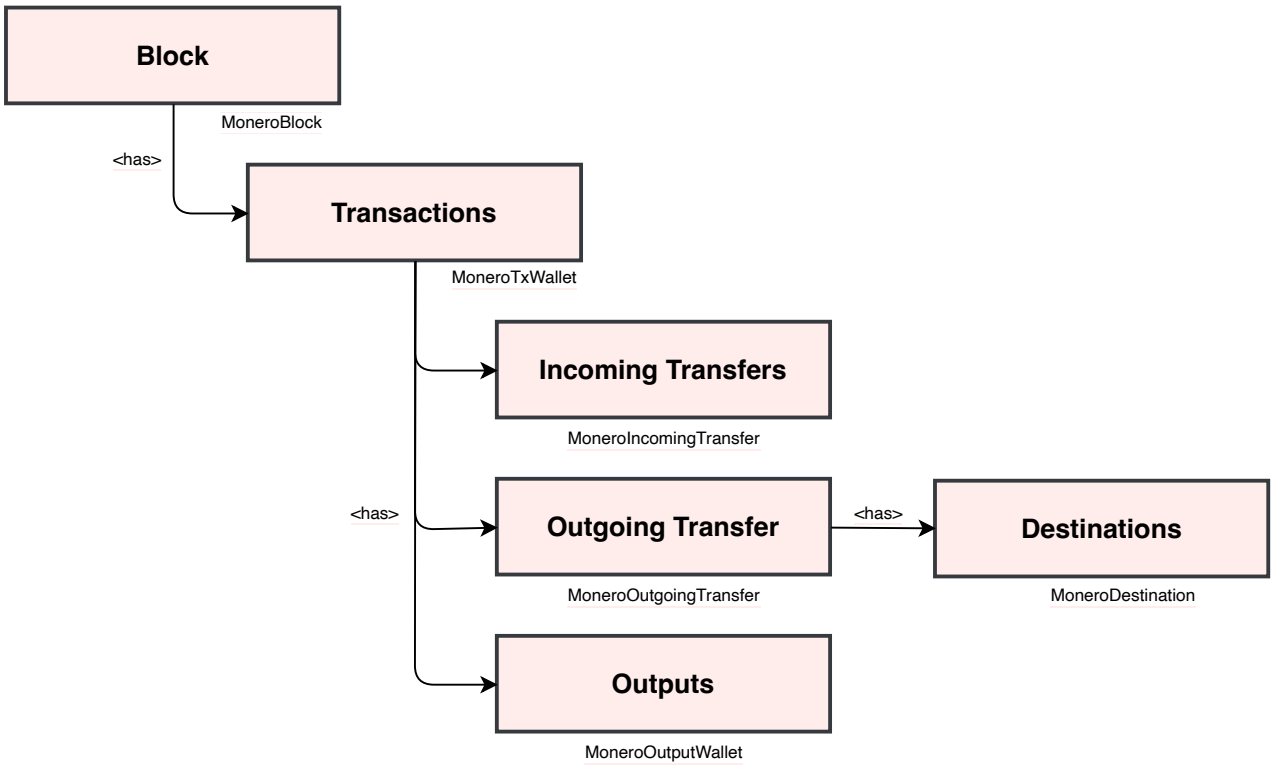
close(bool save=false): void

isClosed(): bool

getIntegratedAddress(string standardAddress=null, string paymentId=null): MoneroIntegratedAddress (deprecated)

decodeIntegratedAddress(string integratedAddress): MoneroIntegratedAddress (deprecated)

Monero Wallet Types 1/4 - Transactions, Transfers, and Outputs



Transactions are part of a block and have incoming transfers and/or an outgoing transfer.

The outgoing transfer may have destination amounts and addresses which are saved locally with the wallet and are not recoverable from the blockchain.

Transactions may also have outputs received by the wallet when requested.

Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs

MoneroTxConfig
address: string
amount: ulong
destinations: MoneroDestination[]
subtract_fee_from: int[]
payment_id: string
priority: MoneroTxPriority
fee: ulong
account_index: uint
subaddress_index: uint
subaddress_indices: uint[]
can_split: bool
relay: bool
note: string
recipient_name: string
below_amount: ulong
sweep_each_subaddress: bool
key_image: string

Configures creating transactions to send funds.

MoneroTxQuery extends MoneroTxWallet
is_outgoing: bool
is_incoming: bool
hashes: string[]
has_payment_id: bool
payment_ids: string[]
height: ulong
min_height: ulong
max_height: ulong
include_outputs: bool
transfer_request: MoneroTransferQuery
output_request: MoneroOutputQuery

Configures a query to get wallet transactions.

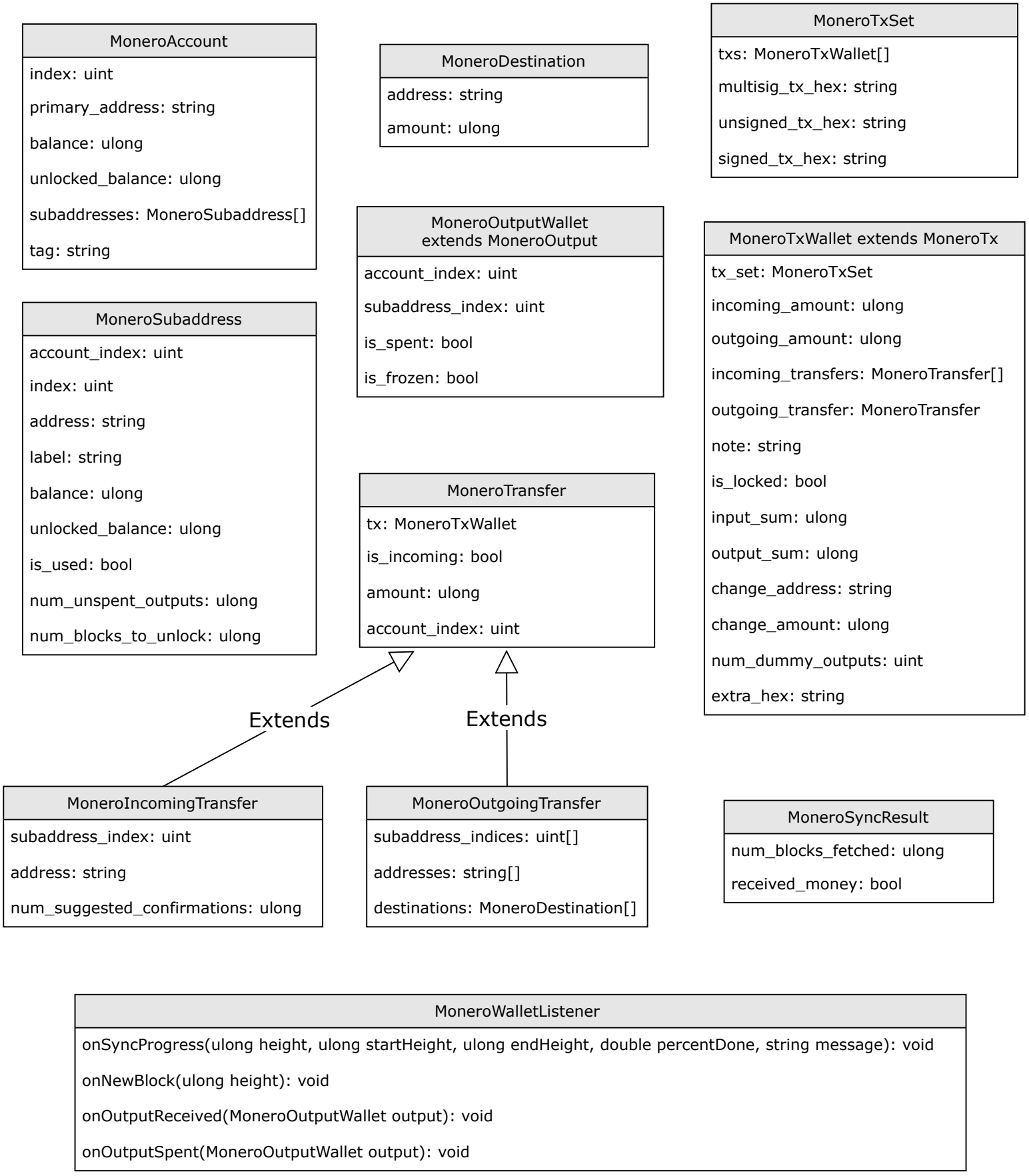
MoneroTransferQuery extends MoneroTransfer
is_incoming: bool
address: string
addresses: string[]
subaddress_index: uint
subaddress_indices: uint[]
destinations: MoneroDestination[]
has_destinations: bool
tx_query: MoneroTxQuery

Configures a query to get wallet transfers.

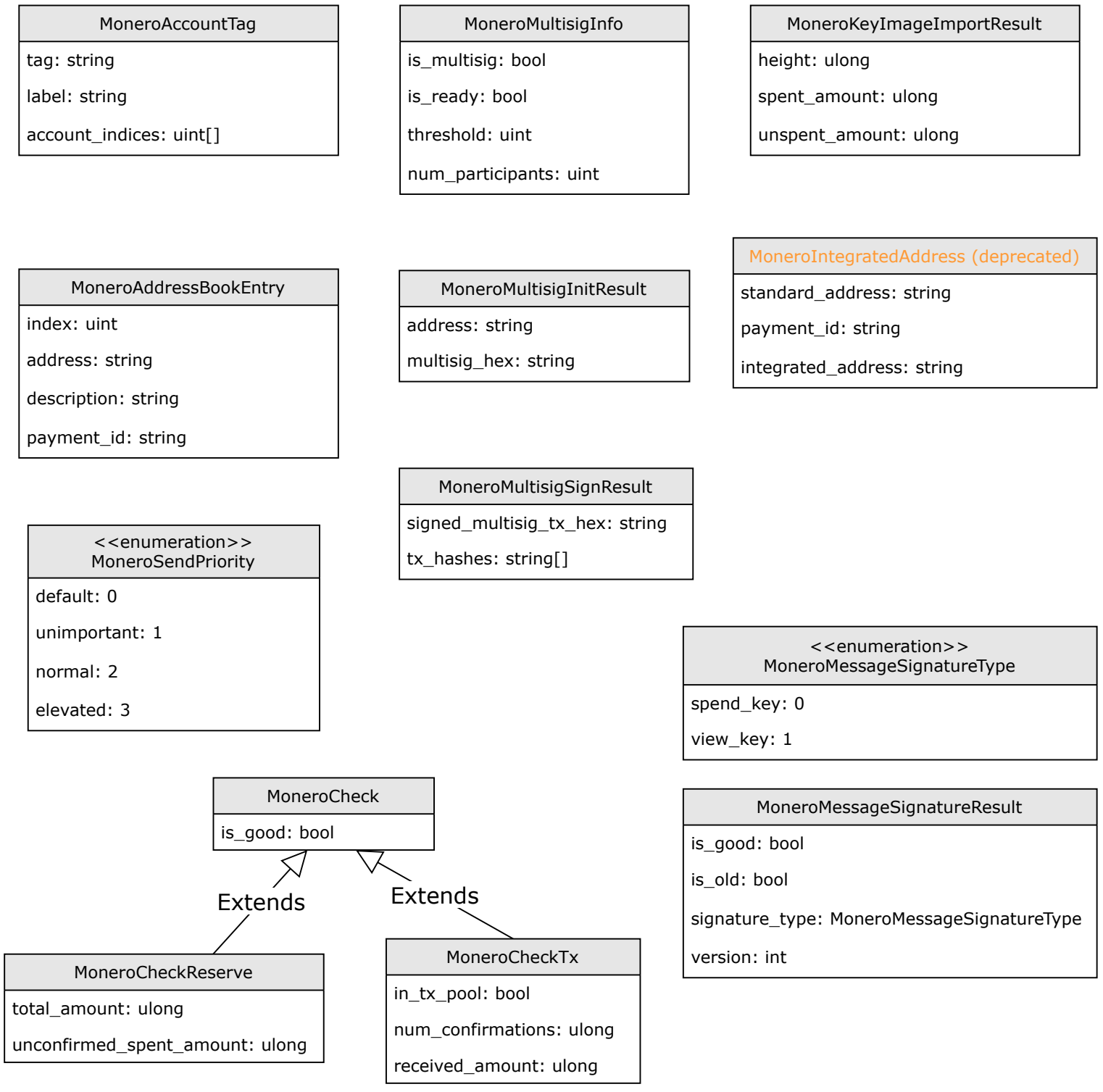
MoneroOutputQuery extends MoneroOutputWallet
subaddress_indices: uint[]
tx_query: MoneroTxQuery
min_amount: ulong
max_amount: ulong

Configures a query to get wallet outputs.

Monero Wallet Types 3/4



Monero Wallet Types 4/4



woodser API 1/1



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